Stable Supply Chains, Falling Growth

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Literature Review

The Decline of Business Dynamism and Aggregate Growth

- **Less dynamism.** Fewer firms enter, fewer jobs are created and destroyed each year, fewer young firms grow fast.
- **More concentration.** Sales concentrate in the largest firms, markups rise, the labour share falls.
- **Less growth.** Productivity growth slows after 2004 while research effort rises.

Literature.

- Slower labour-force growth (Karahan, Pugsley & Şahin 2024; Hopenhayn, Neira & Singhania 2022; Peters & Walsh 2026)
- Superstar firms and intangibles (Autor et al. 2020; Hsieh & Rossi-Hansberg 2023; Aghion et al. 2023; De Ridder 2024)
- Less knowledge diffusion (Akcigit & Ates 2021, 2023)
- Low interest rates (Liu, Mian & Sufi 2022)
- Harder innovation (Bloom et al. 2020; Olmstead-Rumsey forthcoming; Cavenaile, Celik & Tian forthcoming)
- Frictions and regulation (Decker et al. 2020; Davis & Haltiwanger 2014; Grullon, Larkin & Michaely 2019)

▸ Appendix: numbers and sources

Mechanism and Motivating Facts

This Paper

The role of the supply chain on business dynamism and aggregate productivity growth post 2000

A new technology delivers a more volatile supply. As time goes by, the volatility decreases. Since 2000, this volatility difference has been increasing. ▸ Fact 1

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In the supply chain, buyers value the stability of their inputs. ▸ Fact 2

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Mature leaders are harder to take over, so B2B links last longer. ▸ Fact 3

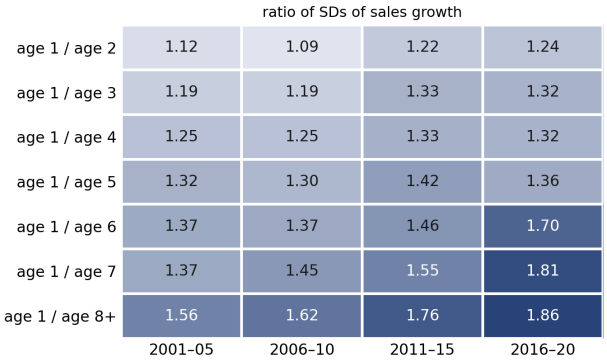
↓

Mature leaders have less incentive to innovate. ▸ Fact 4 ▸ Fact 5

↓

With fewer takeovers and fewer leader innovations, markups rise and growth slows.

Fact 1: Mature Links Deliver a More Stable Supply, and Increasingly So Since 2000



Compustat segment customers, all links, 2001–2020.

◂ Back to This Paper ▸ Appendix: 1981–2024

Fact 2: Links Last Longer at Large, Stable Suppliers

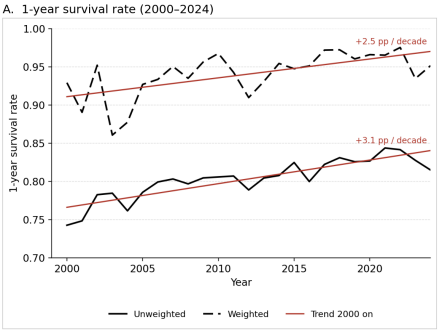
$$\Pr\big(\text{link } sc \text{ alive in } t{+}1 \mid \text{alive in } t\big) = \alpha_{\text{tenure}_{sct}} + \beta^{\top} x_{s,t-1} + \gamma_t + \delta_{\text{SIC}_s} + \varepsilon_{sct}$$

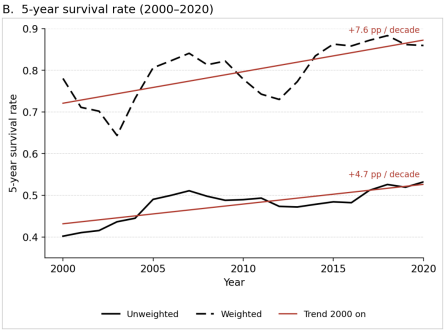
	2000–2024			
	(1)	(2)	(3)	(4)
log employment$_{t-1}$	$+2.46^{***}$			$+1.72^{***}$
	(0.31)			(0.32)
log(1+R&D per worker)$_{t-1}$		-0.58		$+0.26$
		(0.36)		(0.37)
SD of sales growth, $t-5$ to $t-1$			-2.67^{***}	-2.17^{***}
			(0.29)	(0.31)
link-years			67,228	
mean annual survival			0.814	

pp per SD of the lagged supplier characteristic on the annual survival rate; tenure, year and supplier SIC2 fixed effects; standard errors clustered by supplier. 2000–2024 pooled.

◂ Back to This Paper ▸ Appendix: deeper lags

Fact 3: B2B Links Have Lasted Longer Since 2000





US Compustat, links between listed suppliers and their major customers. Survival rate = share of links in year t still reported in $t+1$ (panel A) or in every year through $t+5$ (panel B). Weighted: $S_t^w = \sum_i \left(s_{it} / \sum_j s_{jt} \right) a_{it}$, with s_{it} link i's sales in t and $a_{it} = 1$ if it survives. Red line: least-squares trend, slope in percentage points per decade.

◂ Back to This Paper | ▸ Appendix: Belgium, all B2B links | ▸ Appendix: full sample 1978–2024 | ▸ Appendix: by supplier age

Fact 4: Suppliers with Longer Link Duration in the Past Are Now Large in Sales and Light in R&D Intensity

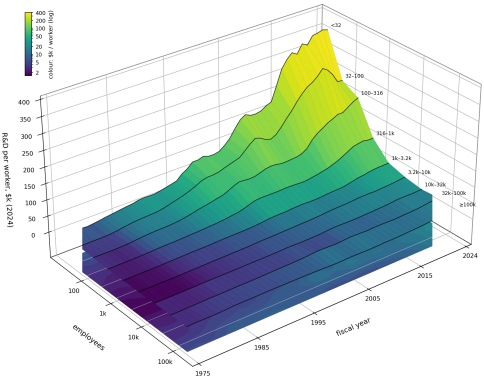
$$y_{st} = b\, T_s^{[t-5,\, t-1]} + \alpha_{\mathrm{age}_{st}} + \gamma_t + \delta_{\mathrm{SIC}_s} + \varepsilon_{st}$$

	2000–2024	
	log sales$_t$ (1)	log(1+R&D per worker)$_t$ (2)
mean link age $T_s^{[t-5,\, t-1]}$, years	$+0.115^{***}$ (0.012)	-0.031^{***} (0.008)
supplier-years		24,385
suppliers		3,656

T = mean age of the supplier's links over $t-5$ to $t-1$ (median 2.1 years); α_{age}: one dummy per year of the firm's age. Standard errors clustered by supplier. Compustat, US non-financial, non-utility suppliers; 2024 dollars.

◂ Back to This Paper ▸ Appendix: sales-weighted

Fact 5: Rising R&D Intensity Dispersion Across Large and Small Firms Since the Mid-1990s



- Since the mid-1990s, R&D intensity has risen drastically in small firms and only modestly in large firms.
- Meanwhile, sales have concentrated in the largest firms. Autor et al. (2020); Grullon, Larkin & Michaely (2019)

Compustat, US non-financial, non-utility firms, 1975–2024. Height = real R&D per employee in the cell (2024 dollars; missing R&D counted as zero). Cell = fiscal year × employment bin (log-spaced), 3-year moving average.

◂ Back to This Paper ▸ Appendix: controlling for industry

A Schumpeterian Model with Quality-Adjusted Quantity Fluctuations

Households

- A representative household derives utility from consuming the final good C_t:

$$U = \int_0^\infty e^{-\rho t} \ln C_t \, dt, \qquad \rho > 0$$

- It supplies labour L inelastically at the wage w_t. The wealth evolution equation is

$$\dot{A}_t = r_t A_t + w_t L - C_t$$

Final Good Production

- A continuum of industries, each supplying one intermediate good. Two firms compete in each and one produces.
- Production takes two steps at every instant t.
  - Step 1: firms post prices p_{jft}. The final-good producer picks a supplier and orders y_{jft}. The supplier buys labour n_{jft} and material X_{jft} to fill the order.
  - Step 2: the shock $\epsilon_{jft} \sim \mathcal{N}\left(-\frac{1}{2}\sigma_{jft}^2, \sigma_{jft}^2\right)$, $\mathbb{E}\, e^{\epsilon_{jft}} = 1$, is realised, and the supplier delivers

$$\underbrace{e^{\epsilon_{jft}}}_{\text{shock}} \times \underbrace{z_{jft}\, n_{jft}^{1-\alpha} X_{jft}^{\alpha}}_{\text{order } y_{jft}}$$

- A competitive final-good producer combines the intermediate goods with a Cobb–Douglas technology. Shocks are independent, so by the LLN Y_t is not random:

$$Y_t = \exp\left(\int_0^1 \ln\left(e^{\epsilon_{jt}} y_{jt}\right) dj\right) \overset{\text{LLN}}{=} \exp\left(\int_0^1 \left(\ln y_{jt} - \sigma_{jt}^2/2\right) dj\right)$$

▸ Appendix: derivation

- The final good is consumed or used as the material input X_{jft} of intermediate firms,

$$Y_t = C_t + \int_0^1 X_{jt}\, dj$$

Effective Price and Limit Pricing within Industries

- Two firms per industry, $f = 1, 2$, each with a posted price p_{jft} and a volatility σ_{jft}. In each industry the final-good producer picks the firm with the lowest effective price, $\tilde{p}_{jft} \equiv p_{jft}\, e^{\sigma_{jft}^2/2}$.
- Market leader limit pricing over the effective price. The leader produces an order at unit cost c_{jt}, the follower at $e^{g_{jt}} c_{jt}$, with g_{jt} the technology difference. The leader's product is new or mature, $s_{jt} \in \{N, M\}$, the follower's is new, and $\sigma_N > \sigma_M$:

$$\underbrace{p_{jt}\, e^{\sigma_{s_{jt}}^2/2}}_{\text{leader's effective price}} = \underbrace{e^{g_{jt}} c_{jt}\, e^{\sigma_N^2/2}}_{\text{follower's effective price at cost}}$$

- The markup and the profit follow, with $\Delta \equiv (\sigma_N^2 - \sigma_M^2)/2$,

$$\mu(g, s) \equiv \frac{p_{jt}}{c_{jt}} = e^{g}\, e^{\Delta \cdot \mathbf{1}[s=M]}, \qquad \pi_t(g, s) = \left(1 - \frac{1}{\mu(g, s)}\right) Y_t$$

Innovation and Successful Taking Over

- Innovation comes with a hazard rate based on R&D effort.
- An innovation builds on the leader's technology z_{jt} and raises it to $z_{jt}\, e^{\lambda}$, with a random step $\lambda \sim \mathrm{Exp}(\theta)$.
- An innovation by the leader, the follower or an entrant succeeds only if its effective price is below the current producer's,

$$\lambda > \Delta \cdot \mathbf{1}[s_{jt} = M]$$
$$p_s \equiv \Pr(\text{adopt} \mid s) = e^{-\theta \Delta \cdot \mathbf{1}[s=M]}$$

Dynamic Problem of Firms: The Leader

- The leader's state is its technology difference from the follower, g, and whether its product is new or mature, $s \in \{N, M\}$.
- A firm becomes the leader only by overtaking the other firm, and its product matures only after that, so $g > 0$.
- The HJB of the leader is

$$
\begin{aligned}
rV_t(g,s) - \dot{V}_t(g,s) = \max_{x_{g,s}} \Bigg\{ & \underbrace{\pi_t(g,s) - c_t(x_{g,s})}_{\text{profit net of R\&D cost}} \\
& + \underbrace{x_{g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(g+\lambda, N) - V_t(g,s)\right]}_{\text{own successful innovation}} \\
& + \underbrace{(x_{-g,s} + \tau)\, p_s \left[\mathbb{E}^s_\lambda V_t(-\lambda, N) - V_t(g,s)\right]}_{\text{replaced by the follower or an entrant}} \\
& + \underbrace{m \cdot \mathbf{1}[s = N] \left[V_t(g,M) - V_t(g,N)\right]}_{\text{product matures}} \Bigg\}
\end{aligned}
$$

- R&D cost: $c_t(x) = \frac{\psi}{2} x^2 w_t$.

Dynamic Problem of Firms: The Follower

The follower's state is $(-g, s)$, its difference from the leader and the leader's product state. The HJB of the follower is

$$
\begin{aligned}
rV_t(-g,s) - \dot{V}_t(-g,s) = \max_{x_{-g,s}} \Big\{ & \underbrace{-c_t(x_{-g,s})}_{\text{R\&D cost}} \\
& + \underbrace{x_{-g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(\lambda, N) - V_t(-g,s)\right]}_{\text{own successful innovation}} \\
& + \underbrace{\tau\, p_s \left[0 - V_t(-g,s)\right]}_{\text{exits}} \\
& + \underbrace{x_{g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(-g-\lambda, N) - V_t(-g,s)\right]}_{\text{the leader innovates successfully}} \\
& + \underbrace{m \cdot \mathbf{1}[s = N] \left[V_t(-g, M) - V_t(-g,N)\right]}_{\text{the leader's product matures}} \Big\}
\end{aligned}
$$

Free Entry

- Entry costs ν research workers per innovation. An innovation lands in a random industry, whose product is of type s with probability ω_s.
- Free entry: the rate of entrants' innovations per industry, τ, adjusts until the cost equals the expected value of an innovation,

$$\nu\, w_t = \sum_{s \in \{N,M\}} \omega_s\, p_s\, \mathbb{E}^s_\lambda V_t(\lambda, N)$$

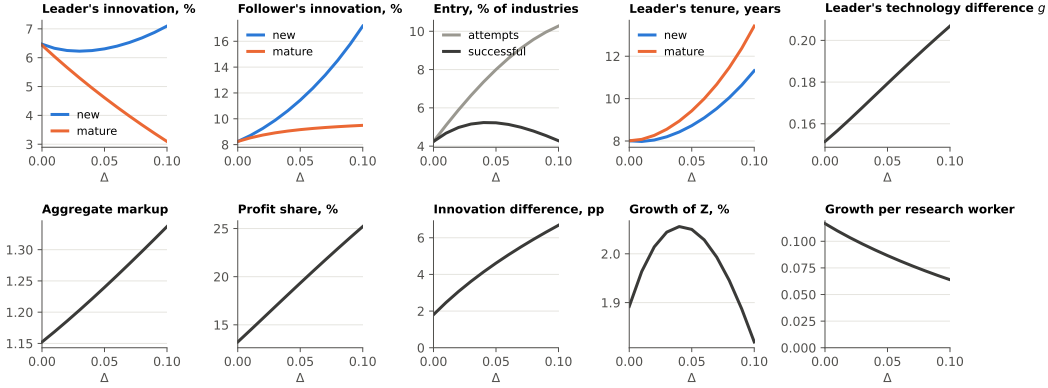
▸ Appendix: balanced growth path, industry structure, aggregation

Quantitative Simulation [Future Work]

Parameters

	Parameter	Value	Set to
ρ	discount rate	0.03	real interest rate 3%
α	material share	0.5	intermediate share of gross output
θ	innovation step $\lambda \sim \mathrm{Exp}(\theta)$	10	mean step 10%
m	maturation rate	1	a new product matures in a year
ψ	R&D cost	14.8	growth of Z 2.05% at $\Delta = 0.05$
ν	entry cost	1.91	entrants' innovation rate 8% per year at $\Delta = 0.05$
Δ	new-product discount	$0 \to 0.10$	the experiment; leader's tenure 8–12 years

Comparative Statics: Δ Increases after 2000



Next Steps

- **Calibration.**
- **Identification of Δ.**
- **Welfare analysis.**

Conclusion

Conclusion

- **Facts.** New supply links deliver a more volatile flow than mature links, and the difference has increased since 2000. Links last longer at large, stable suppliers. B2B links have lasted longer since 2000. Suppliers with lasting links are large in sales and light in R&D intensity. R&D intensity has risen far more in small firms than in large firms.
- **Model.** Schumpeterian growth with quality-adjusted quantity fluctuations. Buyers value a stable supply, so a mature leader is harder to take over and has less incentive to innovate.
- **Result.** A rising volatility difference lengthens leaders' tenure, raises markups and slows growth.

Appendix: Known Facts, Numbers and Sources

Entry	
Firm entry rate: 12.0% → 10.6% → below 8%	Decker et al. (2014)
Employment share of firms aged 5 or less: 18.9% → 13.4%	Decker et al. (2014)
Reallocation	
Job creation 18.9% → 15.8%, destruction 16.1% → 13.4%	Decker et al. (2014)
Firm growth less dispersed; fewer high-growth young firms	Decker et al. (2016)
Employment responds less to the firm's own productivity	Decker et al. (2020)
Market structure	
Sales concentration rises	Autor et al. (2020); Grullon, Larkin & Michaely (2019)
Markup 21% → 61%; profit rate 1% → 8%	De Loecker, Eeckhout & Unger (2020)
Labour share falls, most where concentration rises most	Karabarbounis & Neiman (2014); Autor et al. (2020)
Growth	
Labour productivity growth slows after 2004	Syverson (2017)
Research effort rises, research productivity falls	Bloom et al. (2020)

United States. Entry and job flows: average of the late 1980s → average of 2004–06 (→ Great Recession); markup over marginal cost and profit rate: 1980 → 2016. Decker et al. = Decker, Haltiwanger, Jarmin & Miranda. List after Akcigit & Ates (2021).

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Appendix: Fact 1, 1981–2024

ratio of SDs of sales growth

	1981–85	1986–90	1991–95	1996–00	2001–05	2006–10	2011–15	2016–20	2021–24
age 1 / age 2	1.16	1.15	1.14	1.13	1.12	1.09	1.22	1.24	1.24
age 1 / age 3	1.34	1.20	1.24	1.20	1.19	1.19	1.33	1.32	1.31
age 1 / age 4	1.60	1.26	1.23	1.26	1.25	1.25	1.33	1.32	1.53
age 1 / age 5	1.50	1.49	1.27	1.35	1.32	1.30	1.42	1.36	1.48
age 1 / age 6	1.42	1.41	1.35	1.48	1.37	1.37	1.46	1.70	1.39
age 1 / age 7	1.83	1.30	1.44	1.31	1.37	1.45	1.55	1.81	1.74
age 1 / age 8+	1.66	1.81	1.74	1.60	1.56	1.62	1.76	1.86	2.12

Back to Fact 1

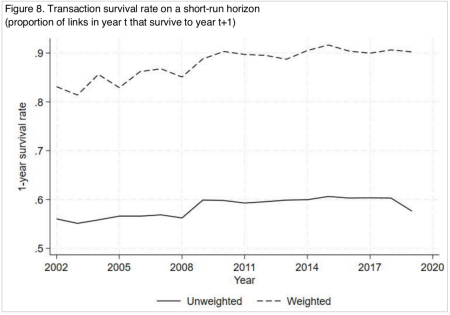
Appendix: Fact 2, Predetermined Characteristics

	baseline, x_{t-1}	supplier alive in $t+5$	x_{t-3}	x_{t-5}
	2000–24	2000–19	2000–24	2000–24
log employment	$+1.72^{***}$	$+1.65^{***}$	$+1.84^{***}$	$+1.52^{***}$
	(0.32)	(0.40)	(0.35)	(0.38)
log(1+R&D per worker)	$+0.26$	$+0.42$	$+0.20$	$+0.08$
	(0.37)	(0.46)	(0.42)	(0.47)
SD of sales growth, five-year window	-2.16^{***}	-2.26^{***}	-1.50^{***}	-1.78^{***}
	(0.31)	(0.44)	(0.34)	(0.37)
link-years	67,228	36,371	51,948	41,508

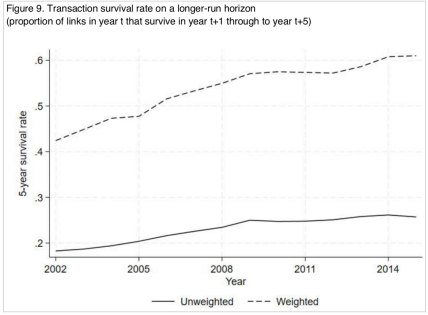
Back to Fact 2

Appendix: Fact 3 in Belgium, All B2B Links

A. 1-year survival rate — NBB WP444 Fig. 8 (2002–2020)



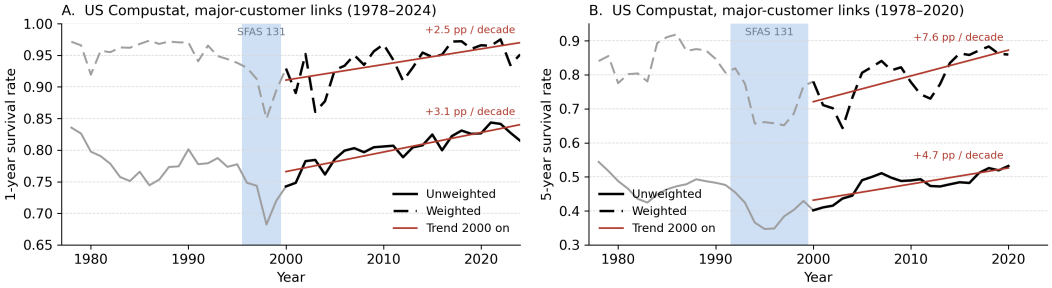
B. 5-year survival rate — NBB WP444 Fig. 9 (2002–2015)



Source: National Bank of Belgium Working Paper 444, Figures 8 and 9. All firm-to-firm links in Belgium, 2002 onward.

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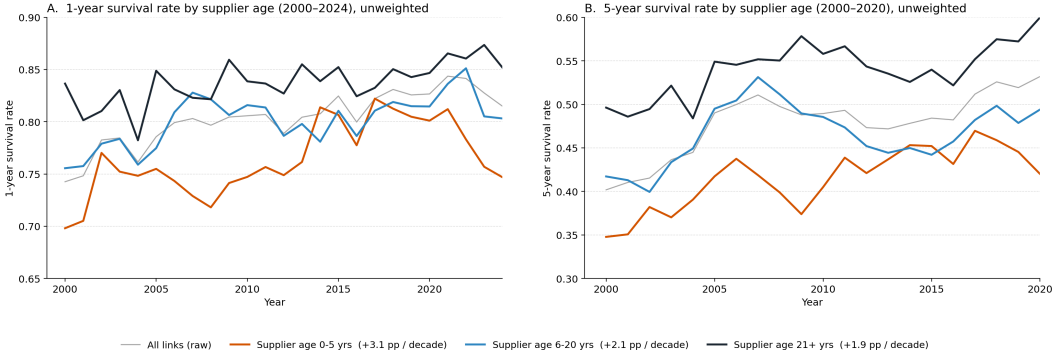
Appendix: Fact 3 on the Full Sample, 1978–2024



US Compustat, supplier–major-customer links. Survival rate = share of year-t links still reported in $t+1$ (A) or through $t+5$ (B); weighted by the link's share of year-t link sales. Blue: years touched by the 1998 SFAS 131 re-collection. Grey: before 2000. Red: trend from 2000, pp per decade.

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Appendix: Fact 3 by Supplier Age



US Compustat, supplier–major-customer links, unweighted. Each line: the survival rate of the year-t links whose supplier has been listed for the stated number of years. Legend: least-squares trend from 2000, pp per decade. Links at suppliers listed 21+ years: 17% of links in 2000–04, 45% in 2020–24.

Back to Fact 3

Appendix: Fact 4 with the Sales-Weighted Link Age

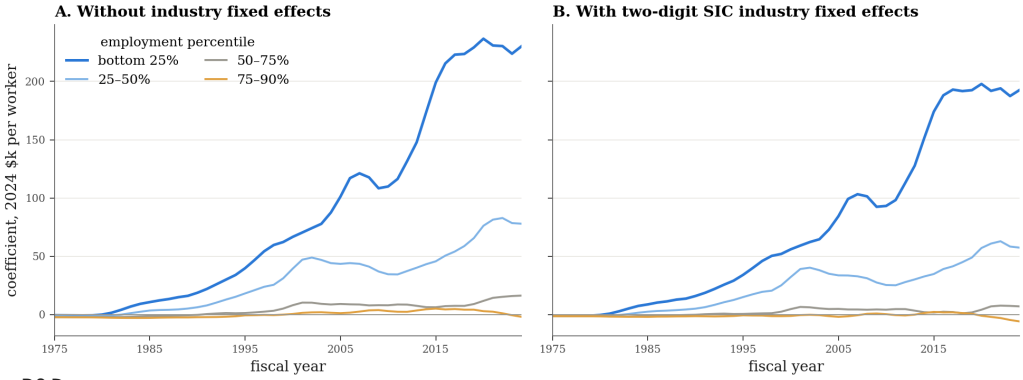
$$y_{st} = b\, T_s^{w,[t-5,\,t-1]} + \alpha_{\mathrm{age}_{st}} + \gamma_t + \delta_{\mathrm{SIC}_s} + \varepsilon_{st}$$

$T_s^{w,[t-5,\,t-1]}$ = the sales-weighted mean age of s's links over $\tau \in [t-5, t-1]$

	2000–2024	
	log sales$_t$	log(1+R&D per worker)$_t$
	(1)	(2)
sales-weighted mean link age $T_s^{w,[t-5,\,t-1]}$, years	$+0.105^{***}$	-0.029^{***}
	(0.011)	(0.007)
supplier-years		22,840
suppliers		3,500

Back to Fact 4

Appendix: Fact 5 Controlling for Industry



$$\frac{\text{R\&D}_{it}}{n_{it}} = \sum_{g \neq \text{top 10\%}} \beta_{gt}\, \mathbf{1}[i \in g] + \alpha_{\text{SIC}_i,\,t} + \varepsilon_{it},$$

weighted least squares, weights n_{it}, year by year

i firm, t fiscal year, n_{it} employees, g employment-percentile group (re-ranked every year), SIC_i the two-digit SIC industry of i; R&D per worker in 2024 $k. Left: without $\alpha_{\text{SIC}_i,\,t}$. Compustat, US non-financial, non-utility firms, 1975–2024; 3-year moving average.

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Appendix: Derivation of the Volatility Discount

- Shocks are independent across industries, so the LLN replaces each ϵ_{jt} by its mean:

$$\int_0^1 \ln\left(e^{\epsilon_{jt}} y_{jt}\right) dj = \int_0^1 \left(\mathbb{E}\left[\epsilon_{jt}\right] + \ln y_{jt}\right) dj$$

- The shock is lognormal with $\mathbb{E}\, e^{\epsilon_{jt}} = 1$, so its log has a negative mean:

$$\mathbb{E}\left[\epsilon_{jt}\right] = -\frac{\sigma_{jt}^2}{2} < 0 = \ln \mathbb{E}\, e^{\epsilon_{jt}}$$

- The certainty-equivalent delivery is $e^{-\sigma_{jt}^2/2}\, y_{jt}$, so

$$Y_t = \exp\left(\int_0^1 \left(\ln y_{jt} - \frac{\sigma_{jt}^2}{2}\right) dj\right)$$

- Intuition: the technology is concave in each good, so a shortfall costs more than an overshoot gains. Under CES with elasticity η the discount is $e^{-\sigma_{jt}^2/(2\eta)}$.

Back to Final Good Production

Appendix: Balanced Growth Path

Let $f(g, s)$ be the density of industries over the leader's state. Flow out of each state equals flow in:

$$
\begin{aligned}
(x_{g,N} + x_{-g,N} + \tau + m)\, f(g, N) = &\int_0^g \left[x_{g',N} f(g', N) + x_{g',M} \mathbf{1}[g - g' > \Delta] f(g', M)\right] \theta e^{-\theta(g-g')}\, dg' \\
&+ \theta e^{-\theta g} \int_0^\infty (x_{-g',N} + \tau) f(g', N)\, dg' \\
&+ \theta e^{-\theta(g-\Delta)} \mathbf{1}[g > \Delta]\, p_M \int_0^\infty (x_{-g',M} + \tau) f(g', M)\, dg'
\end{aligned}
$$

$$
(x_{g,M} + x_{-g,M} + \tau)\, p_M\, f(g, M) = m\, f(g, N)
$$

A balanced growth path is a set of values, innovation rates, τ and a density that satisfy the two HJBs, free entry, labour-market clearing and these conditions, with $V_t \propto w_t \propto Y_t$ and $r - g = \rho$.

Back to the model

Appendix: Steady-State Industry Structure

- Shares of industries by the leader's product; average innovation rates of leaders and of followers:

$$\omega_s = \int_0^\infty f(g,s)\,dg, \qquad \bar{x}_s = \frac{1}{\omega_s}\int_0^\infty x_{g,s}\,f(g,s)\,dg, \qquad \bar{x}_s^F = \frac{1}{\omega_s}\int_0^\infty x_{-g,s}\,f(g,s)\,dg$$

- Flow out of the mature state equals flow in:

$$(\bar{x}_M + \bar{x}_M^F + \tau)\,p_M\,\omega_M = m\,\omega_N, \qquad \omega_N + \omega_M = 1 \quad\Longrightarrow\quad \omega_M = \frac{m}{m + (\bar{x}_M + \bar{x}_M^F + \tau)\,p_M}$$

- Replacement rate and tenure of a leader:

$$\bar{h} = \omega_N(\bar{x}_N^F + \tau) + \omega_M(\bar{x}_M^F + \tau)\,p_M, \qquad \text{tenure} = 1/\bar{h}$$

- Growth: $1/\theta$ per successful step, Δ more when a mature product is replaced, by a challenger or by its own producer:

$$g_Z = \Big(\frac{1}{\theta} + \Delta\Big)\omega_M\big(\bar{x}_M + \bar{x}_M^F + \tau\big)p_M + \frac{1}{\theta}\,\omega_N\big(\bar{x}_N + \bar{x}_N^F + \tau\big)$$

Parameters of the Δ experiment, $\Delta = 0.05$: $\omega_M = 0.88$, $\bar{x}_N = 6.3\%$, $\bar{x}_M = 4.6\%$, $\bar{x}_N^F = 11.4\%$, $\bar{x}_M^F = 9.2\%$, $\tau = 8.0\%$, tenure 8.7 years, $g_Z = 2.05\%$.

Back to the model

Appendix: Aggregation

Let $\Lambda \equiv \int_0^1 \mu_{jt}^{-1}\, dj$ be the cost share of sales.

The final-good producer spends Y_t on each industry. Firms buy material input $X_t = \int_0^1 \alpha\, c_{jt} y_{jt}\, dj = \alpha \Lambda\, Y_t$, so consumption, which is GDP, is

$$C_t = (1 - \alpha\Lambda)\, Y_t$$

Workers produce or do research, $L = L_P + L_R$, and the production labour market clears: $w_t L_P = (1 - \alpha)\, \Lambda\, Y_t$. The numeraire condition gives the wage:

$$w_t = \left[\alpha^{\alpha}(1-\alpha)^{1-\alpha}\, \frac{Z_t\, \bar{a}}{\mathcal{M}}\right]^{\frac{1}{1-\alpha}}, \qquad Z_t \equiv e^{\int_0^1 \ln z_{jt}\, dj}, \quad \mathcal{M} \equiv e^{\int_0^1 \ln \mu_{jt}\, dj}, \quad \bar{a} \equiv e^{-\int_0^1 \sigma_{jt}^2/2\, dj}$$

The profit share of sales is $1 - \Lambda$. On a balanced growth path w_t and Y_t grow at $g = g_Z/(1-\alpha)$.

Back to the model